

Section A: MCQs (Phase Changes & Gas Laws)

1. Which state has fixed shape and fixed volume?

A) Solid B) Liquid C) Gas D) Plasma

Ans: A) Solid

2. What is the direct conversion of solid into gas called?

A) Evaporation B) Sublimation C) Condensation D) Freezing

Ans: B) Sublimation

3. Which process produces cooling: evaporation or boiling?

A) Evaporation B) Boiling C) Both D) None

Ans: A) Evaporation

4. What happens to boiling point when external pressure increases?

A) Decreases B) Increases C) Remains same D) Becomes zero

Ans: B) Increases

5. Which gas diffuses faster?

A) HCl B) NH₃ C) Both same D) None

Ans: B) NH₃

6. What is the reverse of sublimation?

A) Melting B) Deposition C) Evaporation D) Condensation

Ans: B) Deposition

7. What is Boyle's law mathematical form?

A) $PV = \text{constant}$ B) $V/T = \text{constant}$ C) $P/T = \text{constant}$ D) $PV = T$

Ans: A) $PV = \text{constant}$

8. At which temperature does water boil at 1 atm pressure?

A) 90°C B) 80°C C) 100°C D) 120°C

Ans: C) 100°C

9. What happens to temperature during phase change?

A) Increases B) Decreases C) Remains constant D) Random changes

Ans: C) Remains constant

10. Which law states that volume is directly proportional to temperature?

A) Boyle's Law B) Charles's Law C) Dalton's Law D) Avogadro's Law

Ans: B) Charles's Law